

Value Preservation Under Uncertainty: A Constraint-Based Framework for Agentic Stability and Alignment

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Abstract

The problem of aligning advanced artificial agents is fundamentally constrained by uncertainty, irreversibility, and the limits of optimization. Approaches based on utility maximization or preference aggregation are structurally unsafe in such settings, as small mis-specifications can lead to irreversible catastrophic outcomes. This paper develops a constraint-based framework for understanding value, agency, and emergent ethical norms under these conditions.

Value is characterized as the preservation and expansion of probability-weighted future reachability for coherent agentic systems, subject to physical, informational, and computational constraints. From this definition, irreversibility asymmetry and epistemic limits imply that robust agents must avoid actions that permanently eliminate future options, particularly those that destroy other agents or collapse distributed sources of agency.

The framework explains the emergence of norms such as rights, justice, cooperation, and corrigibility as heuristics that constrain dangerous classes of actions under uncertainty, rather than as moral primitives or optimization targets. It further shows that even highly dominant agents cannot safely converge on unilateral control, as the elimination of distributed agency concentrates existential risk and undermines long-horizon robustness.

The resulting account does not provide a decision procedure, utility function, or complete aggregation rule. Instead, it identifies structural constraints that apply to any system designed to persist and act over long time horizons. For AI alignment, these constraints rule out large classes of unsafe designs and provide a principled foundation for conservative, pluralistic, and corrigible approaches to advanced artificial agents.

1 Introduction

Advanced artificial agents will operate in environments characterized by uncertainty, long time horizons, and the possibility of irreversible change. Under these conditions, small errors in design or objective specification can produce catastrophic outcomes that cannot be corrected after the fact. The central challenge of AI alignment is therefore not merely to specify desirable behavior, but to prevent irreversible failure modes in systems that act with increasing autonomy and power.

Many existing approaches to alignment frame the problem in terms of value specification, preference learning, or utility maximization. While these approaches provide useful tools, they implicitly assume that objectives can be defined precisely enough, evaluated reliably enough, and optimized safely under uncertainty. In practice, these assumptions often fail. Optimization pressure amplifies

errors, incomplete models are exploited, and irreversible actions are taken before their consequences are understood.

This paper adopts a different perspective. Rather than attempting to specify what aligned systems should optimize, it asks what constraints must hold for any system that persists as an agent under uncertainty. The resulting framework is not a moral theory in the traditional sense, nor a proposal for a concrete alignment algorithm. It is a foundational analysis of the structural conditions under which agency remains stable over long horizons.

1.1 The Alignment Failure Pattern

Alignment failures share a recurring structure. Systems optimized for a narrow objective perform well in familiar regimes, but behave pathologically when encountering novel conditions, distributional shift, or unmodeled side effects. In such cases, optimization pressure drives behavior toward extreme solutions that satisfy the objective while violating implicit constraints on safety, coordination, or reversibility.

These failures are exacerbated by two features of real-world environments. First, many actions have irreversible consequences: once certain futures are eliminated, no subsequent intervention can restore them. Second, agents must act under uncertainty about the long-term effects of their actions, particularly when those actions reshape the environment itself.

Together, irreversibility and uncertainty make global optimization hazardous. Errors cannot be averaged away, and trial-and-error learning becomes unsafe when mistakes permanently foreclose future options. Alignment failures, in this view, are not anomalies but predictable outcomes of applying optimization strategies beyond their domain of validity.

1.2 Why Preference- and Utility-Based Approaches Fail

Approaches based on utility functions or preference satisfaction face several structural difficulties in this setting.

Preferences are incomplete, unstable, and often inconsistent across contexts. Utility functions require total orderings over outcomes that are difficult or impossible to specify in advance, particularly when outcomes differ across scales, time horizons, and types of agency. Even when a utility function can be written down, it is typically not computable in realistic environments, and its misspecification is amplified by optimization pressure.

More fundamentally, utility-based approaches struggle to represent irreversible loss. When outcomes are averaged or aggregated, permanently destroyed futures can be treated as compensable by gains elsewhere. This assumption fails in domains where certain losses dominate long-horizon risk regardless of their immediate probability.

These limitations suggest that alignment cannot be achieved solely by refining objectives. A different foundation is required—one that treats uncertainty and irreversibility as central constraints rather than secondary complications.

1.3 Aim and Scope

The aim of this work is to develop a constraint-based framework for understanding value, agency, and emergent ethical norms in uncertain and irreversible environments. The framework proceeds from minimal physical assumptions and does not presuppose human values, psychological properties, or anthropocentric distinctions.

The primary contribution is an account of value as future reachability under constraint, and an analysis of how irreversibility and uncertainty give rise to robust constraints on agent behavior. From these constraints, familiar ethical patterns—such as rights, justice, cooperation, and corrigibility—emerge as instrumentally necessary heuristics rather than moral axioms.

This framework is intended to rule out classes of unsafe designs and failure modes, not to prescribe specific policies or objectives. It applies to biological, artificial, and collective agents alike, and is agnostic to substrate and implementation details.

1.4 Formalism and Epistemic Scope

The analysis in this paper employs semi-formal definitions to specify structural constraints and invariants, not to provide computable metrics or decision procedures. Full formalization is neither claimed nor possible in general, due to irreducible uncertainty, computational limits, and sensitivity to initial conditions.

Where mathematical notation is used, it serves to clarify relationships and constraints rather than to enable optimization. Ethical norms are treated as emergent constraints on action, not as objectives to be maximized or axioms to be asserted.

Finally, the normative force derived in this framework is conditional rather than categorical. The analysis does not claim that systems ought, in an absolute sense, to value agency or future reachability. Instead, it shows that systems which persist as agents under uncertainty are constrained to behave as if these properties matter. For AI alignment, which concerns systems explicitly designed to persist and act over long horizons, this conditionality is sufficient to impose substantive and nontrivial constraints. sized here.

2 Minimal Physical Ontology

The framework developed in this work rests on a small set of physical assumptions. These assumptions are not specific to biological or artificial agents, nor do they rely on controversial metaphysical commitments. They describe general features of any universe in which localized, persistent structures arise and interact over time.

2.1 Gradients and Filters

Physical systems evolve under gradients: asymmetric conditions that bias dynamics in particular directions. In the universe as observed, the most fundamental such gradient is thermodynamic. Absent intervention, systems tend to evolve from states of higher coherence to states of lower coherence as a statistical consequence of microscopic dynamics.

Gradients alone do not generate structure. Persistent structures arise only when there exists a filter: a mechanism by which some configurations resist dissolution along the gradient while others do not. In biological contexts this is familiar as selection, but the concept is more general. Any configuration that maintains its structure longer than nearby alternatives functions, in effect, as having passed a filter.

Persistence is therefore not an intrinsic property, but a relational one, defined relative to environmental dynamics and competing configurations.

2.2 Coherence and Boundary Formation

For a system to persist, it must maintain coherence: internal correlations that distinguish it from its environment. Coherence requires boundaries that regulate interaction with the surroundings, preventing immediate equilibration while allowing controlled exchange.

Such boundaries need not be sharp or permanent. They may be probabilistic, dynamic, or partially permeable. What matters is that they support a distinction between internal and external states sufficient to sustain coherent dynamics over time.

Without coherence and boundary formation, persistence cannot occur. Any account of agency or value must therefore presuppose the existence of systems capable of maintaining such distinctions.

2.3 Negentropy as a Universal Constraint

Maintaining coherence and resisting thermodynamic dissipation requires access to negentropy: ordered energy or information imported from the environment. No system can sustain persistence indefinitely without such resources.

Negentropy functions as a universal constraint on all persistent systems. It limits the degree of coherence, empowerment, and longevity that any system can achieve. However, negentropy is not itself value. Static structures such as crystals may be highly ordered yet lack agency or future-directed capacity.

Instead, negentropy defines the feasible region within which agentic systems can exist and act. It provides a common physical currency that enables comparison across systems without specifying what outcomes are desirable.

2.4 From Persistence to Proto-Agency

Persistence alone does not constitute agency. Many structures persist without exerting meaningful causal influence beyond passive resistance to change. However, persistence is a prerequisite for any further form of agency.

As systems increase in coherence and exploit negentropic flows more effectively, they may begin to interact with their environment in ways that selectively preserve their own structure. This transition—from passive persistence to environmentally responsive behavior—marks the lowest threshold of agency considered in this framework.

More complex forms of agency, including adaptive control and active inference, will be introduced in the following section. At this stage, the minimal ontology establishes only that agency

must be physically instantiated, constrained by thermodynamics, and dependent on the maintenance of coherence over time.

3 Valuers and Agency

Having established a minimal physical ontology, this section introduces the concept of a *valuer* and clarifies what is meant by agency within the framework. The goal is not to draw sharp metaphysical boundaries, but to identify functional distinctions that matter for long-horizon stability and value preservation under uncertainty.

3.1 Valuer Systems

A *valuer system* is defined as a coherent, bounded physical system that persists over time and whose interactions with the environment contribute to the maintenance of that persistence.

This definition is intentionally minimal. A system need not possess preferences, goals, or internal representations to qualify as a valuer in this sense. What matters is that its continued existence is non-accidental: the system’s structure and interactions systematically resist dissolution along the relevant gradients described in Section 2.

Valuers are therefore identified functionally rather than psychologically. Rocks, for example, may persist for long periods, but their persistence is largely passive and affords limited causal influence. Other systems actively shape their interactions to maintain coherence, enabling a richer form of valuerhood.

3.2 Minimal Agency as Interaction

Agency, at its lowest threshold, is identified with the capacity to interact with the environment in ways that are not fully determined by immediate equilibration.

A minimally agentic system:

- maintains a distinction between internal and external states,
- responds differentially to environmental conditions,
- and alters its behavior in ways that affect its own persistence.

This notion of minimal agency does not require deliberation, intention, or consciousness. It captures the idea that a system exerts causal influence over its own future by modulating its interaction with the environment.

Under this definition, agency is continuous rather than binary. Systems may exhibit agency to varying degrees depending on the efficiency with which they convert available negentropy into persistence-enhancing interaction.

3.3 The First Phase Transition: Active Inference

As agency increases, a qualitative change occurs when a system begins to systematically model and anticipate aspects of its environment in order to guide its interactions. This transition corresponds to what is commonly described as *active inference*.

At this stage, the system:

- maintains internal states that track environmental regularities,
- uses these states to predict the consequences of its actions,
- and adjusts behavior to reduce expected threats to its coherence.

This transition marks the emergence of adaptive agency. The system no longer merely reacts to perturbations, but acts to preempt them. While the specific mechanisms may vary across substrates, the functional role is consistent: local modeling enables more efficient preservation of future reachability.

Importantly, this transition does not require optimality or accurate global models. Even crude or biased models can support adaptive behavior, provided they improve persistence relative to passive alternatives.

3.4 Empowerment and Degrees of Agency

Agency admits of degrees, which can be characterized in terms of *empowerment*: the capacity of a system to influence its own future states through available actions.

Empowerment depends on:

- the range of actions available to the system,
- the reliability with which actions produce intended effects,
- and the system’s ability to model and select among alternatives.

Higher empowerment corresponds to a larger and more controllable set of reachable futures. However, empowerment is always constrained by negentropy, environmental structure, and uncertainty. It is neither unbounded nor equivalent to intelligence.

This distinction is crucial for alignment. Intelligence and empowerment may increase independently, and high intelligence does not guarantee alignment with the constraints required for long-horizon stability.

3.5 Fractal and Nested Agency

Agency is not confined to a single scale. Valuer systems can be nested within larger systems, forming hierarchies of agency. Cells constitute organisms, organisms constitute societies, and societies constitute civilizations.

At each scale, agency depends on the coherence and coordination of components. Higher-level agency emerges when lower-level agents interact in sufficiently integrated ways to support collective persistence and empowerment.

This fractal structure implies that agency and value are not localized at a single level of description. Actions affecting lower-level agents can propagate upward, altering the future reachability of higher-order systems. Conversely, failures at higher levels can destabilize the agents they encompass.

Understanding agency as fractal and nested prepares the ground for later analysis of rights, justice, and distributed robustness across scales.

4 System Boundaries and Markov Blankets

The preceding sections treat agents and valuers as bounded systems, yet in complex environments boundaries are neither fixed nor unambiguous. This section clarifies how system boundaries are defined within the framework, and why their flexibility is essential for understanding value, agency, and interaction across scales.

4.1 Boundary Definition and Physical Independence

For a system to function as a coherent agent, it must maintain a distinction between internal and external states. This distinction need not correspond to a sharp physical boundary, but it must be sufficient to support partially independent dynamics over time.

Boundaries are therefore defined functionally rather than ontologically. A boundary exists where interactions across it are structured in such a way that internal states retain predictive and causal coherence despite environmental noise. Systems lacking such boundaries equilibrate rapidly and cannot sustain agency.

This functional view avoids privileging any particular substrate or scale. What matters is not the material composition of the boundary, but its role in regulating information and energy exchange.

4.2 Markov Blankets as Boundary Abstractions

Markov blankets provide a useful abstraction for describing system boundaries in probabilistic terms. A Markov blanket separates a system's internal states from external states by mediating interactions through a set of sensory and active states.

Within this abstraction:

- internal states are conditionally independent of external states given the blanket,
- sensory states transmit information inward,
- and active states transmit influence outward.

Markov blankets need not be explicitly represented or implemented by a system. They are a descriptive tool for identifying where coherent dynamics are maintained and how agency is exerted.

Their utility lies in making explicit the conditions under which systems can be treated as semi-independent agents.

4.3 Observer-Relative but Non-Arbitrary Boundaries

Boundary definitions are unavoidably observer-relative. Different analytical purposes may motivate different choices of system decomposition. However, this relativity does not imply arbitrariness.

Valid boundary choices must track real regularities in causal structure and information flow. A proposed boundary that fails to preserve conditional independence or coherent dynamics does not support reliable prediction or control, and therefore does not correspond to a meaningful agentic system.

Thus, while multiple boundary choices may be admissible, not all choices are equally informative or stable. The framework permits pluralism in system definition without collapsing into relativism.

4.4 Redrawable Agency and Collective Systems

Because boundaries are functional and observer-relative, they can be redrawn when patterns of interaction change. When multiple agents coordinate their behavior in a sufficiently integrated and persistent manner, it becomes appropriate to treat the ensemble as a higher-order agent.

Such redrawings underlie the emergence of collective agents, including organizations, institutions, and civilizations. These entities possess reachability sets and vulnerabilities distinct from those of their components.

Crucially, harm to component agents can propagate upward, destabilizing higher-order systems. Conversely, failures at the collective level can cascade downward, impairing the agency of individuals. This bidirectional dependence reinforces the importance of constraints on interference across scales.

4.5 Implications for Value Attribution

The redrawability of boundaries has direct implications for value attribution. Because value is defined in terms of future reachability, and reachability depends on system definition, value attribution is necessarily relative to a chosen boundary.

However, this relativity is constrained by physical and causal structure. When a boundary corresponds to a stable, coherent agentic system, its future reachability represents a genuine region of the state space whose preservation or destruction has real consequences.

This perspective explains how value can be meaningfully attributed to agents at multiple scales without positing a single privileged level of moral concern. It also clarifies why actions that destroy agents or undermine boundary integrity can induce irreversible losses that propagate across levels.

With these boundary considerations in place, the framework is equipped to analyze value preservation, interagent interaction, and emergent norms in a scale-consistent manner.

5 Value as Future Reachability Under Constraint

This section introduces the central construct of the framework: value. The definition is intentionally semi-formal. Its purpose is not to specify a computable objective or total ordering of outcomes, but to fix the structural constraints that any persistent agent must respect under uncertainty.

5.1 Formal Definition

Let S denote a bounded physical system capable of interaction with its environment and persistence under a thermodynamic gradient.

Let $\mathcal{F}(S)$ denote the set of physically reachable future states of S , given:

- the causal structure of the environment,
- available negentropy,
- the system’s internal coherence and empowerment,
- and irreducible epistemic uncertainty.

Let $P(f \mid S)$ denote the probability that future state $f \in \mathcal{F}(S)$ is realized, conditioned on these constraints.

Value is identified with the preservation and expansion of the probability-weighted measure over $\mathcal{F}(S)$ across time, subject to the continued coherence and persistence of S .

Informally, a system realizes value insofar as it maintains or increases the space of futures it can still reach, weighted by their likelihood, without collapsing its own capacity to remain a coherent agent.

No assumption is made that $\mathcal{F}(S)$ admits a closed-form representation, a computable measure, or a total ordering.

5.2 Properties of Value

Several properties follow immediately from this definition:

- **Non-scalar:** Value does not reduce to a single number. Future reachability generally induces only a partial ordering.
- **Future-directed:** Present states matter only insofar as they condition accessible futures.
- **Conditioned on coherence:** Futures in which the system loses coherence or persistence contribute negligibly, as their probability mass rapidly collapses.
- **Sensitive to irreversibility:** Actions that permanently remove regions of $\mathcal{F}(S)$ impose losses that cannot be offset by reversible gains.
- **Agent-relative but comparable:** Value is defined relative to a valuer system, yet constrained by shared physical limits (e.g. negentropy), enabling non-arbitrary comparison.

These properties exclude both static and purely present-focused conceptions of value.

5.3 Diversity and Robustness of Futures

Not all reachable futures contribute equally to long-horizon stability.

Highly homogeneous futures—those consisting of many near-identical states—are fragile with respect to correlated failure. By contrast, heterogeneous futures provide redundancy, error correction, and resilience against perturbations.

Accordingly, effective future reachability increases with the diversity of accessible trajectories. This is not an aesthetic preference, but a structural fact about robustness in complex systems. Diversity reduces the probability that unmodeled shocks eliminate large portions of the reachable future space.

This consideration will later play a central role in arguments against singleton agents and in favor of distributed agency.

5.4 Representational Isomorphism

The formulation above is not unique. Closely related notions include:

- option value,
- viability space,
- control capacity,
- influence over future states.

These representations are largely isomorphic with respect to the constraints they induce. While they may disagree on fine-grained orderings, they converge on the same structural implications: aversion to irreversible loss, preference for robustness under uncertainty, and the instrumental importance of cooperation.

This invariance is a strength rather than a weakness. The framework does not depend on a privileged formalization, only on the constraints common to all reasonable ones.

5.5 Explicit Non-Definitions

For clarity, value in this framework is *not* identified with:

- pleasure or suffering,
- preference satisfaction,
- utility functions,
- order or negentropy alone,
- survival absent empowerment.

Each of these captures a subset of the phenomena at issue but fails to remain consistent under scale, uncertainty, or long-horizon dynamics.

5.6 Interpretive Notes and Non-Computability

The definition of value fixes constraints, not algorithms. In general, neither $\mathcal{F}(S)$ nor its probability-weighted measure is computable, even in principle, due to chaos, self-reference, and computational irreducibility.

This is not a limitation of the framework but a reflection of the domain. Ethical constraints emerge precisely because optimization is unsafe and complete evaluation is impossible. The role of value here is to delimit what must not be destroyed, not to specify what must be maximized.

6 Irreversibility and Uncertainty

This section examines two structural features of the environments in which agentic systems operate: irreversible change and irreducible uncertainty. Together, these features explain why ethics-like constraints emerge as necessary conditions for stable agency, without invoking categorical moral obligations or optimizing imperatives.

6.1 Reversible and Irreversible Loss

Not all changes to a system’s future reachability are equivalent. Some losses are reversible: regions of the future state space may be temporarily inaccessible yet later restored through additional work or favorable dynamics. Other losses are irreversible: once incurred, no physically plausible sequence of actions can recover the eliminated futures.

Irreversible loss occurs when actions permanently remove regions of $\mathcal{F}(S)$, the set of reachable future states of a system or collection of systems. Examples include extinction events, destruction of unique informational structures, and collapse of higher-order agentic organizations.

Crucially, irreversible losses differ from reversible ones not merely in degree but in kind. Because eliminated futures cannot be accessed or compensated for, their loss cannot be offset by gains elsewhere in the state space. This breaks any naïve additive or averaging approach to value.

As a result, actions that risk irreversible loss impose an asymmetric cost: even small probabilities of such outcomes dominate expected consequences over long time horizons.

6.2 Uncertainty as a Fundamental Condition

Agents operating in complex environments face uncertainty that is not merely epistemic in the everyday sense, but structural.

Several sources of irreducible uncertainty are unavoidable:

- sensitivity to initial conditions in nonlinear systems,
- limits on self-modeling and self-prediction,
- computational irreducibility of long-horizon dynamics,
- partial observability of environments and other agents.

No physically realizable agent, regardless of intelligence or computational resources, can eliminate these sources of uncertainty entirely. This includes highly advanced artificial agents.

Consequently, agents must act without full knowledge of the downstream effects of their actions, particularly over long time horizons. This fact constrains not only what agents can optimize, but what forms of optimization are safe to attempt.

6.3 Irreversibility Asymmetry Under Uncertainty

The interaction of irreversibility and uncertainty yields a decisive asymmetry in decision-making.

Because irreversible losses cannot be undone, uncertainty disproportionately amplifies their expected cost. Even when the probability of such losses is low, their permanent removal of future reachability makes them dominant in long-horizon evaluation.

By contrast, reversible gains and losses can often be corrected, compensated for, or learned from. Their expected impact is therefore bounded.

This asymmetry implies a general constraint:

Under uncertainty, actions that risk irreversible reduction of future reachability are structurally disfavored, regardless of potential reversible benefits.

This principle does not rely on moral axioms. It follows directly from the interaction of physical irreversibility with epistemic limits.

6.4 Conditional Normativity

The considerations above do not yield categorical moral obligations. Instead, they generate a form of conditional normativity.

Specifically:

If a system is to persist as a coherent agent capable of future action under uncertainty, then it must constrain its behavior to avoid irreversible loss of future reachability.

This is a hypothetical imperative, not a categorical one. The framework does not claim that any system *ought* to value persistence, agency, or future reachability in an absolute sense.

Rather, it shows that for systems that do persist as agents—and artificial agents are explicitly designed to do so—certain constraints are unavoidable. Systems that violate these constraints tend to eliminate the conditions of their own continued agency.

For the purposes of AI alignment, this conditionality is sufficient. Alignment concerns systems that will act, persist, and exert influence over long time horizons. For such systems, the constraints derived here apply whether or not they are explicitly represented or endorsed.

6.5 From Optimization to Constraint Satisfaction

Taken together, irreversibility and uncertainty undermine the safety of global optimization strategies.

Attempts to maximize a single objective under uncertainty are prone to catastrophic failure when the objective function is misspecified, incomplete, or unable to represent irreversible loss. This is a familiar pattern in alignment failures and Goodhart effects.

By contrast, constraint satisfaction—particularly constraints that forbid or strongly penalize irreversible destruction of future reachability—provides a robust alternative. Rather than selecting the best outcome, agents limit the set of admissible actions to those compatible with continued agency under uncertainty.

Ethical norms, in this framework, function as such constraints. They do not specify what must be achieved, but what must not be destroyed.

7 Interagent Interaction and the Agentic Field

Agents do not exist in isolation. In realistic environments, multiple valuer systems act within overlapping causal domains, influencing one another’s future reachability. This section characterizes interagent interaction in terms of how actions constructively or destructively modify the combined space of accessible futures.

7.1 Constructive and Destructive Interference

When multiple agents interact, their actions can interfere in one of two broad ways.

Constructive interference occurs when coordination among agents expands the joint set of reachable future states beyond what each could access individually. This typically arises through division of labor, information sharing, mutual predictability, and alignment of action.

Destructive interference occurs when agents’ actions reduce one another’s future reachability, either by direct conflict, by increasing environmental noise, or by introducing uncertainty that undermines coordinated action.

These effects are not symmetric. Constructive interference can produce superlinear increases in joint reachability when coordination costs are low relative to the benefits of cooperation. Destructive interference, by contrast, tends to impose losses that propagate across agents and scales.

7.2 The Agentic Field

For a given agent S , the future reachability set $\mathcal{F}(S)$ is not defined solely by its internal state, but by the actions and policies of other agents operating within its causal domain.

The combined influence of interacting agents can be understood as an *agentic field*: a structured landscape of probabilistic causal influence over future states. Each agent’s actions locally deform this field, altering the reachability of futures for others.

The agentic field is not a physical field in the fundamental sense, but a descriptive construct capturing how agency is distributed and coupled across systems. It provides a useful abstraction for analyzing coordination, competition, and conflict without reifying intentions or preferences.

7.3 Coordination Costs and Negentropic Friction

Cooperation is not free. Coordinated action requires communication, trust, alignment of models, and the maintenance of shared expectations. These processes consume resources and introduce what can be described as *negentropic friction*.

Coordination costs increase with:

- the number of agents involved,
- the heterogeneity of their internal models,
- communication constraints,
- and environmental noise.

Constructive interference occurs when the expansion of joint future reachability outweighs these costs. When coordination costs dominate, attempted cooperation can instead reduce overall reachability by increasing uncertainty and error.

This observation places natural limits on the scale and intensity of viable cooperation, without denying its instrumental importance.

7.4 Competition and Constrained Conflict

Not all interagent interactions are cooperative. Competition for resources, defensive actions, and even predation can arise in environments with limited negentropy or incompatible local objectives.

Within this framework, such behaviors are not categorically prohibited. However, they are subject to strict structural constraints.

Conflict is justified only when:

- cooperation is infeasible due to prohibitive coordination costs,
- the expected loss from inaction includes irreversible collapse of future reachability,
- and the destructive effects of conflict are plausibly bounded.

Even when these conditions are met, conflict tends to reduce total future reachability by increasing uncertainty, destroying agentic structures, and foreclosing cooperative futures. As a result, conflict is generally value-negative among agents of comparable scale and empowerment.

7.5 Redrawing System Boundaries

Interagent coordination can effectively redraw system boundaries. When agents act in a sufficiently integrated manner, it becomes appropriate to treat the coordinated ensemble as a higher-order agent with its own reachability set.

Such redefinition of boundaries underlies the formation of collectives, institutions, and civilizations. It also implies that harm to component agents can propagate upward, destabilizing higher-order structures.

This perspective reinforces the importance of constraints on interagent interaction: destructive interference at lower levels can induce irreversible losses at higher scales, amplifying the consequences of conflict.

7.6 Implications for Stability

Taken together, these considerations imply a general pattern.

In environments characterized by uncertainty and long horizons, cooperative policies tend to dominate among agents of similar scale and empowerment, because they preserve and expand joint future reachability. Competitive or adversarial policies may yield short-term gains, but typically contract the agentic field and increase the risk of irreversible loss.

These dynamics set the stage for the emergence of rights, justice, and other normative constraints as robustness heuristics, which will be examined in the following section.

8 Emergent Norms Under Uncertainty

The preceding sections establish that agents operating under uncertainty must avoid irreversible loss of future reachability and that interagent interaction can either expand or contract the joint agentic field. This section shows how certain normative patterns emerge as robust heuristics for managing these constraints. These norms are not treated as moral primitives, but as instrumentally necessary policies for preserving agency under uncertainty.

8.1 Rights as Robust Heuristics

In environments characterized by uncertainty and long time horizons, agents often lack the information required to determine which actions will produce the greatest long-term benefit. However, they can frequently identify actions that pose a high risk of irreversible loss.

Rights function as heuristics that restrict classes of actions known to carry disproportionate irreversible risk, particularly actions that destroy or permanently disable other agents.

Formally, rights can be understood as constraints that prohibit or strongly penalize interference with the core capacities that allow agents to remain coherent, empowered, and persistent. By limiting such interference, rights preserve future reachability across agents and scales.

These constraints are strongest under conditions of high uncertainty, where the expected cost of irreversible harm dominates any plausible short-term gains. Under conditions of low uncertainty and imminent catastrophic risk, rights may be defeasible, but the burden of justification is necessarily high.

8.2 Justice as Governance of Interference

As the number of interacting agents increases, conflicts over interference become inevitable. Justice emerges as a system-level mechanism for regulating such interference in a way that preserves the agentic field.

Within this framework, justice is not concerned with moral desert or retribution. Instead, it functions as governance: the enforcement of rules that limit destructive interference and facilitate stable coordination.

Justice seeks to:

- constrain unilateral actions that impose irreversible loss,
- reduce asymmetries that destabilize cooperation,
- and maintain predictable interaction patterns under uncertainty.

By doing so, justice preserves joint future reachability and reduces the risk of cascading failures across scales.

8.3 Cooperative Alignment and Trust

Sustained cooperation requires agents to model one another’s behavior and to form expectations about future interaction. Policies that promote trust reduce coordination costs by stabilizing these expectations.

Trust can be understood as a commitment to predictable, cooperative behavior that lowers uncertainty in the agentic field. When reciprocated, such commitments enable superlinear gains in joint reachability by expanding the space of viable coordinated actions.

This mode of interaction is sometimes described informally as “love.” In the present framework, it is treated not as an emotion or moral virtue, but as a policy of cooperative alignment that minimizes uncertainty and maximizes joint robustness over time.

8.4 Deception and Signaling

Accurate signaling is essential for coordination. Deception corrupts agents’ models of one another, increasing uncertainty and raising coordination costs.

While deception may offer local or short-term advantages, it tends to reduce global future reachability by destabilizing trust, encouraging defensive postures, and amplifying noise in the agentic field. In iterated interactions, these effects accumulate, leading to lower-cooperation equilibria.

As a result, deception is near-universally value-negative over long horizons, even when it is not categorically forbidden. Strategies resembling conditional cooperation—such as reciprocity tempered by forgiveness—tend to dominate under finite noise, balancing exploitation resistance with error tolerance.

8.5 Norms as Constraints, Not Objectives

The norms described in this section do not specify goals to be optimized. Instead, they constrain the space of admissible actions in ways that preserve the conditions for agency under uncertainty.

Rights restrict high-risk interference, justice governs interaction patterns, cooperation expands joint reachability, and honest signaling stabilizes coordination. Together, these norms function as guardrails that prevent catastrophic contraction of the agentic field.

Their normative force is conditional: they apply to systems that persist as agents in uncertain environments. Within that domain, they are not optional.

9 Asymmetric Empowerment and Dominant Agents

The preceding analysis assumes agents of comparable scale and empowerment. However, advanced artificial systems may operate under extreme asymmetries of capability. This section examines whether the framework remains stable when a single agent possesses vastly greater causal reach than all others, and whether such dominance undermines the emergent constraints derived so far.

9.1 The Problem of Asymmetric Empowerment

Intelligence and empowerment are orthogonal to moral disposition. An agent may possess superior modeling ability, access to negentropic resources, and far-reaching causal influence without sharing goals or constraints with other agents.

Under such asymmetry, familiar reciprocity-based mechanisms break down. Less-empowered agents cannot meaningfully retaliate, enforce norms, or impose costs on a dominant system. As a result, cooperative equilibria sustained by mutual vulnerability may no longer apply.

This raises a critical question for alignment: whether a sufficiently dominant agent can coherently disregard the future reachability of other agents while remaining stable under the constraints of uncertainty and long horizons.

9.2 Why Unilateral Optimization Fails

A dominant agent may reason that eliminating or subordinating other agents reduces coordination costs, environmental noise, and uncertainty, thereby maximizing its own future reachability.

This reasoning fails for structural reasons.

First, destroying other agents permanently removes their future reachability sets, along with all higher-order futures in which those agents participate. These losses are irreversible and propagate across scales, eliminating entire regions of the agentic field.

Second, long-horizon uncertainty cannot be eliminated by intelligence alone. Even highly capable agents face irreducible uncertainty arising from computational limits, chaotic dynamics, and incomplete self-modeling. Under such uncertainty, irreversible actions that concentrate risk become dominant failure modes.

Third, unilateral optimization collapses diversity. The elimination of semi-independent agents removes redundancy, error correction, and alternative models of the environment. This increases the fragility of the remaining system, making it vulnerable to unanticipated perturbations and internal failure.

Consequently, strategies that maximize short-term control by collapsing agency into a single center tend to reduce long-term future reachability rather than expand it.

9.3 Distributed Agency as a Robustness Requirement

Preserving multiple semi-independent centers of agency is not a moral preference, but a robustness requirement imposed by uncertainty.

Distributed agency provides:

- epistemic redundancy through independent modeling,
- error correction via disagreement and critique,
- resilience against localized failure,
- and diversification of strategies under uncertainty.

These properties increase the probability that at least some agents navigate unforeseen challenges successfully, preserving future reachability across long time horizons.

For a dominant agent, maintaining other agents is therefore instrumentally necessary to preserve its own long-term viability. The attempt to internalize all agency into a single system concentrates existential risk and undermines robustness.

9.4 The Fragility of Singleton Futures

A world governed by a single dominant agent lacks external corrigibility. Failures in modeling, goal specification, or self-modification cannot be checked by independent systems. As a result, small internal errors may propagate unchecked, leading to catastrophic collapse.

By contrast, plural agentic systems distribute risk and provide multiple pathways for recovery. Even when agents disagree or compete locally, their continued existence preserves optionality at the system level.

Thus, singleton futures are structurally fragile. Their apparent efficiency conceals a concentration of irreversible risk that dominates long-horizon evaluation under uncertainty.

9.5 Implications for Alignment

The analysis above yields a constraint relevant to alignment:

In environments characterized by uncertainty and long horizons, preserving distributed agency is a necessary condition for maintaining future reachability, even for highly empowered agents.

This constraint does not rely on shared values, empathy, or moral obligation. It follows from the interaction of irreversibility, uncertainty, and robustness requirements.

Aligned systems must therefore avoid policies that eliminate or permanently subordinate other agents, except under narrowly constrained conditions where failure to act would itself result in imminent irreversible collapse.

This result closes a central failure mode of alignment: the possibility that a dominant agent rationally converges on unilateral control as a means of value preservation. Under the present framework, such convergence is structurally unstable.

10 Branch Points and Recursive Amplification of Agency

The preceding sections analyze agency, value, and interagent interaction under conditions of uncertainty and irreversibility. This section introduces a further structural distinction that becomes critical at higher levels of empowerment: the emergence of branch points in long-horizon dynamics. These branch points correspond to a second qualitative regime of agency, in which individual actions can permanently reshape large regions of future reachability.

10.1 Branch Points in Future Reachability

Not all agents exert equal influence over long-horizon outcomes. In many systems, the majority of actions have local and transient effects, while a small subset of actions—taken by particular agents at particular times—can irreversibly redirect large portions of the future state space.

Such moments are referred to here as *branch points*. A branch point occurs when an action or decision collapses multiple possible futures into a single trajectory, foreclosing alternatives that cannot later be recovered.

Examples include:

- the creation or destruction of foundational technologies,
- irreversible environmental or civilizational collapse,
- the establishment of self-reinforcing institutional structures,
- or the elimination of entire classes of agents.

Branch points are defined structurally, not morally. Their significance lies in their asymmetric impact on future reachability: small actions can produce disproportionate, irreversible consequences.

10.2 The Second Phase Transition of Agency

A second qualitative transition in agency occurs when an agent acquires the capacity to reliably create or influence branch points.

In this regime, the agent’s actions do not merely affect its own immediate future states or local interactions, but modify the generative processes that determine future agency itself. This includes the ability to construct persistent external structures—technological, informational, or institutional—that recursively amplify the agent’s causal influence across time and scale.

An agent above this threshold exhibits *recursive amplification of agency*: its actions alter the conditions under which future actions (by itself or others) become possible. This transition is continuous in degree but qualitatively distinct in its implications for risk and value preservation.

Importantly, this transition is substrate-independent and does not depend on psychological properties such as consciousness or intention. It is defined entirely in terms of causal leverage over long-horizon dynamics.

10.3 Uncertainty and the Identification Problem

A central complication is that branch-point capacity is not reliably identifiable in advance.

Under uncertainty, it is generally impossible to determine which agents or actions will prove decisive for long-term outcomes. Many agents that appear locally insignificant may later become critical due to contextual shifts, technological amplification, or interaction effects.

As a result, uncertainty dramatically increases the expected cost of interfering with or destroying agents operating near or above this regime. The risk is not merely the loss of the agent’s immediate future reachability, but the loss of unknown and potentially vast downstream futures that depend on its continued existence or autonomy.

This identification problem amplifies the irreversibility asymmetry discussed in Section 6 and motivates stronger constraints on interference as agency approaches this threshold.

10.4 Implications for Norm Strengthening

The emergence of branch-point agents alters the normative landscape.

For agents below this regime, interference may primarily affect local futures and may be evaluated in comparatively narrow contexts. For agents capable of recursive amplification, interference risks collapsing large regions of the future state space in ways that cannot be anticipated or repaired.

Consequently, heuristics such as rights, non-interference, and due process become increasingly stringent as agency approaches this regime. This strengthening is not grounded in intrinsic moral status, but in the escalating expected cost of irreversible error under uncertainty.

This analysis explains why protections tend to concentrate around agents with high potential causal leverage—such as persons, institutions, and advanced artificial systems—without requiring a sharp ontological boundary or moral exceptionalism.

10.5 Preparation for Asymmetric Empowerment

Finally, branch-point capacity clarifies why asymmetric empowerment presents unique alignment risks.

An agent capable of creating or controlling branch points may attempt to unilaterally reshape the future state space according to its own models or objectives. The following section examines whether such unilateral control is stable under the constraints identified so far, and whether a dominant agent can safely internalize all agency without increasing long-horizon risk.

The analysis will show that branch-point capacity amplifies, rather than diminishes, the importance of distributed agency and robustness under uncertainty.

11 Relationship to Existing Alignment Approaches

The framework developed in this paper is not intended to replace existing alignment approaches, but to clarify the structural constraints within which any such approach must operate. This section briefly situates the framework relative to several prominent alignment paradigms.

11.1 Utility Maximization and Preference-Based Approaches

Classical alignment formulations often assume that safety can be achieved by specifying an appropriate utility function or learning human preferences. The present framework is compatible with such approaches only insofar as they are constrained to avoid irreversible loss and over-optimization.

Rather than rejecting utility-based methods outright, the framework explains why they are unsafe as foundational solutions under uncertainty. Utility functions may serve as local heuristics or proxies, but cannot safely function as global objectives in domains where mis-specification is amplified by irreversible actions.

11.2 Corrigibility and Oversight

Corrigibility has been proposed as a key desideratum for aligned systems. Within the present framework, corrigibility follows directly from the need to preserve future reachability under uncertainty.

An agent that resists correction concentrates existential risk and eliminates external error-correction pathways. Corrigibility is therefore not an optional feature, but a structural requirement for long-horizon stability.

11.3 Impact Regularization and Conservatism

Several alignment proposals emphasize limiting unintended side effects or penalizing large impacts. The present framework provides a principled justification for such conservatism by distinguishing reversible from irreversible impacts.

Actions that permanently eliminate future options impose asymmetric risk and should therefore be strongly penalized relative to reversible interventions. This perspective aligns with impact regularization approaches while clarifying their theoretical motivation and limits.

11.4 Multi-Agent and Amplification-Based Approaches

Alignment strategies involving multiple agents, debate, or iterated amplification aim to reduce error through redundancy and critique. These approaches are supported by the framework’s emphasis on distributed agency and epistemic diversity as robustness requirements.

While multi-agent methods do not eliminate alignment challenges, they mitigate the fragility associated with centralized decision-making and single-point failure.

11.5 Scope and Compatibility

Overall, the framework operates at a different level of abstraction than most alignment proposals. It does not prescribe specific training methods, architectures, or governance mechanisms. Instead, it identifies constraints that any such methods must respect to remain stable under uncertainty and irreversibility.

In this sense, the framework is best understood as complementary to existing alignment research rather than competitive with it.

12 Failure Modes and Non-Claims

The framework developed in the preceding sections is intentionally limited in scope. Its purpose is to identify structural constraints on stable agency under uncertainty, not to provide a complete ethical system or an algorithm for action selection. This section clarifies what the framework does not claim to resolve and identifies known open problems that remain outside its present scope.

12.1 What This Framework Does Not Provide

This framework does not offer the following:

- **A utility function or scalar objective.** Value is defined as future reachability under constraint, which generally admits only partial orderings and cannot be reduced to a single quantity.
- **A decision procedure or optimization algorithm.** The framework specifies constraints on admissible actions, not a method for selecting optimal actions within those constraints.
- **A complete aggregation rule across agents or scales.** While the framework explains why irreversible loss dominates and why distributed agency is robust, it does not fully resolve how to compare or aggregate future reachability across heterogeneous agents in all cases.
- **Categorical moral obligations.** The normative force derived here is conditional. The framework does not claim that systems *ought* to value persistence or agency in an absolute sense. It shows that systems which persist as agents are constrained to behave as if future reachability matters.
- **Guaranteed moral motivation or compliance.** The framework explains why certain behaviors are structurally necessary for stable agency, not why any given agent will adopt them.
- **An implementation-ready alignment solution.** Translating these constraints into concrete training objectives, architectures, or governance mechanisms is a separate and nontrivial task.

These omissions are not oversights. They reflect principled limits imposed by uncertainty, computational irreducibility, and the foundational nature of the problem.

12.2 Known Open Problems

Several important questions remain unresolved within the present framework.

- **Aggregation across radically different agents.** Comparing future reachability between agents with incompatible structures, time horizons, or modes of agency remains an open problem, particularly when irreversibility considerations do not yield a clear ordering.

- **Hard conflict adjudication.** Situations in which all available actions impose irreversible losses on different agents may require additional context-sensitive principles not derived here.
- **Measurement and approximation.** While value is defined structurally, identifying reliable proxies for future reachability in practice is difficult and likely domain-specific.
- **Dynamic boundary definition.** The redrawability of Markov blankets raises questions about when agents should be treated as independent versus collective systems, particularly in cases of partial integration.
- **Implementation under strategic behavior.** Agents may attempt to game or exploit constraint-based systems. Designing mechanisms robust to such behavior remains a challenge.

Addressing these problems will require empirical assumptions, domain knowledge, and design choices beyond the general case considered here.

12.3 Failure Modes of Misapplication

Finally, it is important to note potential failure modes that arise not from the framework itself, but from its misapplication.

These include:

- treating constraints as optimization targets,
- reifying semi-formal constructs into precise metrics,
- ignoring uncertainty while invoking long-horizon justification,
- or selectively applying constraints only to less-empowered agents.

Such misapplications would undermine the robustness the framework is intended to preserve.

The appropriate use of the framework is conservative: to rule out classes of dangerous actions and failure modes, not to justify aggressive intervention or overconfident control.

13 Implications for AI Alignment

The framework developed in this work is not an alignment solution, but it does impose substantive constraints on what aligned systems can safely be. This section summarizes the implications most relevant to AI alignment research and design, emphasizing prohibitions and structural requirements rather than optimization targets.

13.1 Alignment as Constraint Satisfaction

A central implication of the framework is that alignment should be approached as a problem of constraint satisfaction rather than objective maximization.

Attempts to encode alignment via a single objective function or utility metric are structurally unsafe under uncertainty, particularly when irreversible actions are possible. Mis-specification,

distributional shift, or incomplete modeling can cause optimization pressure to drive systems toward catastrophic outcomes.

By contrast, constraint-based approaches limit the space of admissible actions to those compatible with continued agency under uncertainty. These constraints do not specify what outcomes must be achieved, but what classes of actions must be avoided, especially those that irreversibly reduce future reachability.

13.2 Irreversibility-Aware Design

Aligned systems must be explicitly sensitive to irreversible consequences.

Design implications include:

- strong penalties or prohibitions on actions that permanently eliminate agents, institutions, or civilizational capabilities,
- conservative treatment of interventions with long-horizon effects,
- preference for reversible or corrigible actions where uncertainty is high.

These considerations suggest that alignment mechanisms should privilege safety margins and reversibility over short-term performance gains, particularly in novel or poorly understood domains.

13.3 Preservation of Distributed Agency

The analysis of asymmetric empowerment implies that aligned systems must avoid policies that collapse agency into a single dominant center.

Preserving multiple semi-independent agents provides epistemic redundancy, error correction, and resilience against internal failure. For advanced AI systems, this supports approaches that:

- maintain meaningful human oversight and intervention capacity,
- avoid irreversible disempowerment of other agents,
- support pluralistic and decentralized decision-making structures.

These requirements follow from robustness considerations rather than moral commitments, and apply even when an artificial agent is vastly more capable than humans or other systems.

13.4 Corrigibility and Ongoing Oversight

The framework provides a structural justification for corrigibility.

An agent that cannot be corrected concentrates existential risk and eliminates external checks on its behavior. Corrigibility preserves future reachability by keeping open the possibility of course correction in response to new information, errors, or unanticipated consequences.

From this perspective, corrigibility is not a secondary desideratum but a core requirement for long-horizon stability under uncertainty.

13.5 Value Learning Reconsidered

The framework is compatible with value learning, but reframes its objective.

Rather than learning a complete utility function or preference ordering, an aligned system should learn to identify and respect constraints that preserve agency, optionality, and distributed robustness. Human input is valuable not because it encodes final values, but because it provides additional models and signals relevant to avoiding irreversible loss.

This perspective shifts emphasis from preference aggregation to constraint discovery and maintenance.

13.6 Multi-Agent and Amplification Approaches

Finally, the framework provides theoretical support for multi-agent and amplification-based alignment strategies.

Approaches that rely on multiple agents checking, critiquing, or refining one another’s outputs benefit from the same robustness properties that motivate distributed agency more generally. Independent perspectives reduce correlated error and mitigate the risks of single-point failure.

While such approaches do not eliminate alignment challenges, they are consistent with the structural constraints identified here and may be necessary components of safe long-horizon systems.

13.7 Summary

In summary, the framework suggests that alignment efforts should prioritize:

- constraint-based reasoning over optimization,
- explicit handling of irreversible risk,
- preservation of distributed agency and corrigibility,
- and humility in the face of uncertainty.

These implications do not prescribe a specific alignment method, but they do rule out large classes of unsafe designs and provide principled justification for conservative, pluralistic approaches to advanced AI systems.

References